audio-technica®

DUAL MAGNET PHONOGRAPH CARTRIDGES

INTRODUCTION

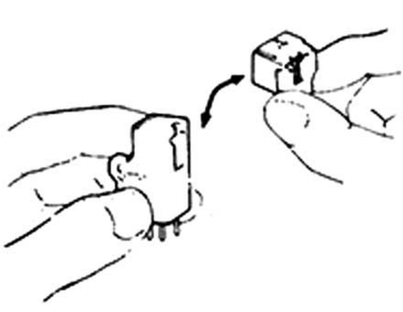
Your new Audio-Technica cartridge is a significant achievement in precision manufacturing. Completely hand assembled, tolerances are held to just thousandths of an inch. Audio-Technica has specialized in precision transducers for over a decade, building an international reputation for accuracy and uniformity. The Dual Magnet Series of phono cartridges is the latest refinement of this tradition of craftsmanship and advanced design.

PRELIMINARY INSTALLATION

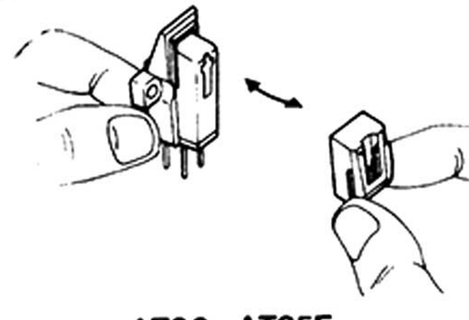
Before mounting the cartridge, the stylus assembly should be removed for safekeeping. Hold the cartridge in one hand, grasp the plastic needle housing with thumb and forefinger, and pull directly away from the cartridge. The stylus assembly comes away at a slight rearward angle, as indicated by the arrow in Figure 1. Place the stylus assembly out of harm's way until installation is completed.

Using the mounting hardware supplied either with the cartridge or with the turntable, mount the cartridge according to the recommendations of the turntable manufacturer. Hardware should be tightened down slightly, so that the cartridge position does not change, but avoid excessive force which may strip plastic threads or warp delicate parts out of position. After the cartridge is mounted, replace the stylus assembly briefly. The stylus assembly should click into place when fully seated on the cartridge. Check visually to make sure that there is no mechanical interference with mounting hardware. Again remove the stylus assembly for safekeeping until electrical connections are completed.

Fig. 1



AT90 AT91 AT94E



AT93 AT95E

ELECTRICAL CONNECTIONS

Four terminals are provided at the rear of the cartridge – an output and ground connection for each stereo channel. For proper performance of your system, the correct wiring must be carried through to the amplifier inputs. Connect the turntable wiring to the cartridge, observing the lead designation furnished with the turntable. The cartridge pins are color coded, using a well-established standard in the turntable industry. DO NOT SOLDER TO THE CARTRIDGE TERMINALS. USE SLIP-ON LUGS TO WHICH TONE ARM WIRES HAVE BEEN SOLDERED BEFORE PLACING ON CARTRIDGE TERMINALS. Heat applied directly to the cartridge terminals will damage the internal cartridge connections.

For monaural operation, the left and right signal terminals should be connected to the monaural signal lead, and the left and right ground terminals should be connected to the ground lead. With these connections, vertical output from the cartridge is cancelled, thus reducing record noise which is primarily in the vertical direction.

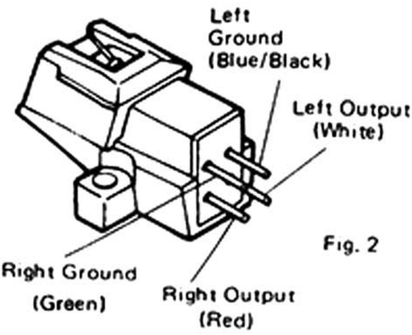


Fig. 2

FINAL INSTALLATION

Re-install stylus assembly. Following the procedure recommended by the turntable or tone arm manufacturer, carefully set tracking force within the appropriate range shown in the specification table. This is a critical adjustment, since tracking force higher or lower than recommended, will result in increased distortion and decreased record life. Similarly, the tone arm's adjustments, if any, for stylus overhang and anti-skating must be set properly if maximum benefit is to be obtained from your cartridge.

After all tone arm adjustments have been made, the stylus may remain installed in the cartridge. However, the plastic stylus guard should be used to protect against accidental damage, as at any time when the turntable is not in use.

REPLACEMENT STYLI

Audio-Technica grain-oriented diamond styli are carefully selected and polished for minimum wear to themselves and to your records. However, after an extended period of play, even the finest diamond stylus becomes worn. For this reason, it is best to have the stylus inspected annually, or whenever it appears that wear or damage may be a problem. The dealer from whom you purchased this cartridge is equipped and competent to inspect and evaluate the stylus.

Should a replacement stylus be required, accept only a genuine Audio-Technica replacement stylus. Only genuine Audio-Technica styli with the genuine A-T circle monogram, as on your original stylus, will provide the same level of performance found in the original cartridge stylus.

FULL ONE-YEAR WARRANTY

Audio-Technica cartridges and styli are guaranteed for one year from date of purchase to be free of defects in materials and workmanship. In event of any defect, product will be repaired without charge or, at our option, replaced with a new product. Warranty excludes normal stylus wear or failure due to improper use with consequent damage. For instructions on return for repair of Audio-Technica products, please contact your local Audio-Technica retailer.

CELLULES PHONOLECTRICES A DOUBLE AIMANT (DUAL MAGNET)

INTRODUCTION

Cette nouvelle cellule Audio-Technica marque une étape dans le domaine de la fabrication de précision. Entièrement montée à la main, ses tolérances sont de l'ordre de quelques centièmes de millimètre. Depuis plus de dix ans, les transducteurs de précision sont la spécialité d'Audio-Technica qui a ainsi acquis une réputation internationale pour la précision et la fiabilité de ses produits. Les cellules à double aimant représentent le raffinement le plus récent de cette tradition d'artisanat et de conception d'avant-garde.

Technical Data	AT95E	AT94E	AT93	AT91	AT90
Frequency Response (Hz)	20–20,000	20–20,000	20–20,000	20–20,000	20–20,000
Output Voltage (mV) (1 KHz, 5 cm/sec)	3.5	3.5	3.5	3.5	3.5
Channel Separation (dB) (1 KHz)	More than 20	More than 20	More than 20	More than 20	More than 20
Channel Balance (dB) (1 KHz)	Within 2.0	Within 2.5	Within 2.0	Within 2.5	Within 3.0
Tracking Force (g) (Nominal Force 2.0 g)	1.5–2.5	1.5–2.5	1.5–2.5	1.5–2.5	1.5–2.5
Coil Impedance (Ω) (1 KHz)	2,800	2,800	2,800	2,800	2,800
Coil DC Resistance (Ω)	410	410	410	410	410
Coil Inductance (mH)	400	480	400	480	480
Dynamic Compliance ($\times 10^{-6}$ cm/dyne)	6.5	6.5	6.5	6.5	6.5
Static Compliance ($\times 10^{-6}$ cm/dyne)	20	20	20	20	20
Load Resistance (KΩ)	47	47	47	47	47
Load Capacitance (pF)	100–200	100–200	100–200	100–200	100–200
Stylus (mil)	0.4×0.7	0.4×0.7	0.6	0.6	0.6
Vertical Tracking Angle	20°	24°	20°	24°	24°
Weight (g)	5.7	5.0	5.7	5.0	5.0
Dimension (m/m)	L30.5×W17.0×H17.2	L31.0×W16.3×H17.0	L30.5×W17.0×H17.2	L31.0×W16.3×H17.0	L31.0×W16.3×H17.0
Cantilever	Aluminum	Carbon	Aluminum	Carbon	Carbon
Replacement Stylus	ATN–95E	ATN–94E	ATN–93	ATN–91	ATN–90

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